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DEPARTMENT OF INFORMATICS,

BIOENGINEERING, ROBOTICS AND SYSTEM ENGINEERING

MECHANICAL DESIGN METHODS IN ROBOTICS

REPORT

Design for a gripper used to pick up the front upper motor mount

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Summary

Requirements

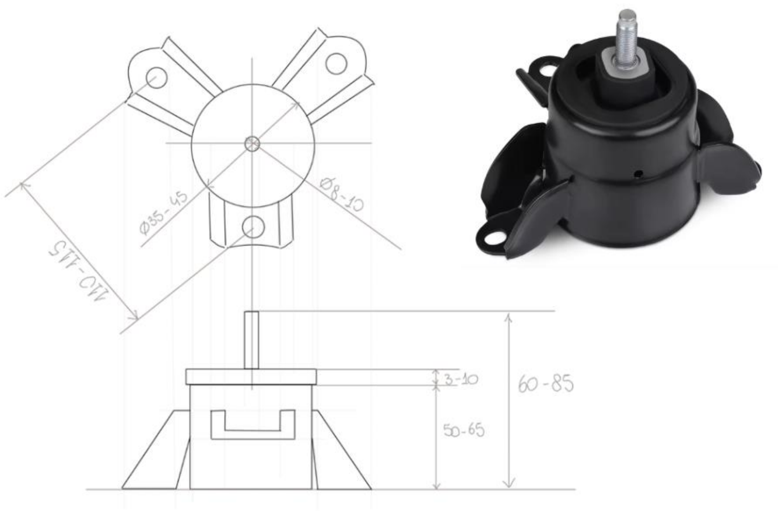
The objective of this report is to analyse the target component (top motor mount) to be grasped and to achieve functional and geometric requirements for the design of a suitable robotic gripper.

The design process considers the geometric characterisation of the motor top mount, identifying graspable areas and a set of dimensions. Based on these parameters, a set of grasping requirements is formulated, which will guide the selection of the most appropriate gripper architecture.

Pick-Up Component:

Front upper motor mount

The target component is a **front upper motor mount**. It consists of a central cylindrical body, a top threaded screw, and three radial flanges, which are further defined as “petals” for convenience.



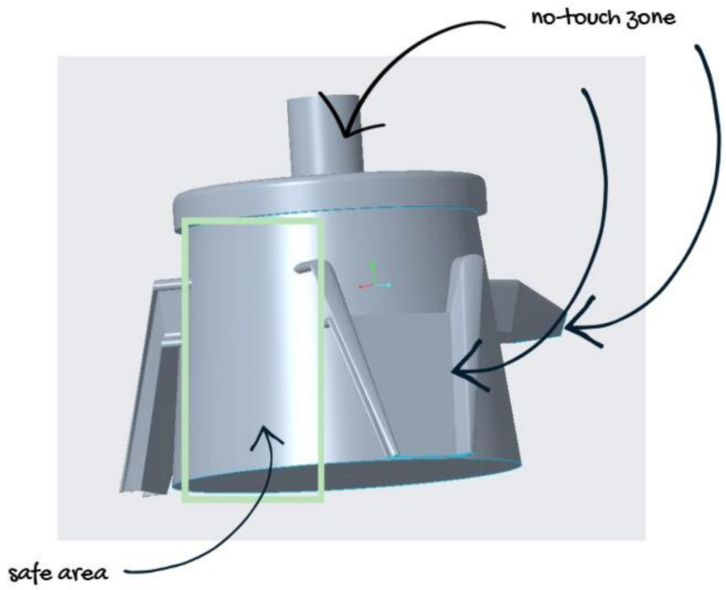
Regarding the dimensions, we consider the central cylinder has a **diameter** between 35 mm and 45 mm, while the total **height** of the piece reaches a maximum of 85 mm (inclusive of the upper screw). Since the component is relatively small and light, we can consider its **mass** negligible for the general structural design of the gripper.

To ensure a safe pick-up process, we have identified specific "restricted zones" that the gripper must avoid to prevent damage:

- **Threaded Screw:** This area is excluded because the gripping force could easily damage the delicate threads.
- **Bolt Sections:** These areas where bolts secure the mount are out of bounds due to the low ductility of the material in those sections.

- **Painted Surfaces:** The top part of the casing is painted; a hard metallic gripper could chip the paint. This requirement suggests the potential use of a gripper with a "softer touch" or specific contact pads.

We can conclude, the only safe and reliable area for grasping is the **lower cylindrical surface**. This can be better seen in Figure below. To ensure stability and avoid the three flanges, a **three-finger gripper architecture** has been selected.



Design assumptions

The final design considers the following assumptions:

- The angle between the 'petals' has constant angles of 120°.
- The change in diameter and height of the component is considered.
- The lateral surface of the component has a conical geometry.

For testing purposes, the following CAD was introduced to test the actual grip of the designed gripper. The dimensions used <u>considerer the biggest size of component</u>.

Gripper Design Options

The objective is to design a gripper integrated with a **Jodell motor**. The Jodell motor features two symmetrical sliders for precise linear movement.

To ensure the gripper is adaptable it will have the following design considerations:

- Adapt to the allowable safe touching areas which are determined by the components 'petals'. This ensures the gripper can handle asymmetrical designs.
- Adapt to the change of diameter and height of the component.

Therefore, we are prioritizing a **reconfigurable set-up** where finger positions can be manually adjusted to avoid collisions.

To ensure the gripper meets industry standards, we considered the following in the design consideration:

- Ease of manufacturing
- Minimal packaging and assembly
- Conduct maintenance in an efficient manner
- Choice of material and design should be appropriate to handle the stresses applied

All the options considered in this report are based on **three-finger gripper architecture**.

The following sections describe the continuous development process. Each design represents a step forward in our research to find a gripper that meets all functional requirements while respecting the project's strict deadline.

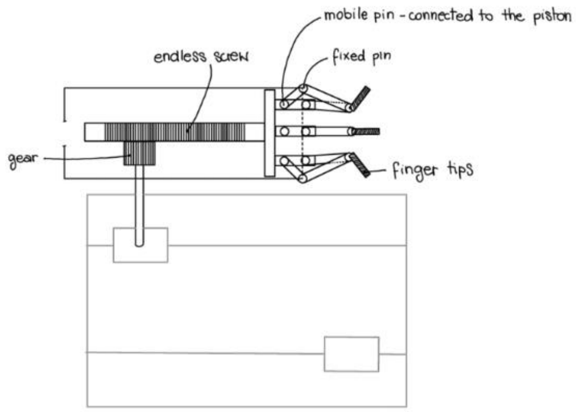
Each option is evaluated against:

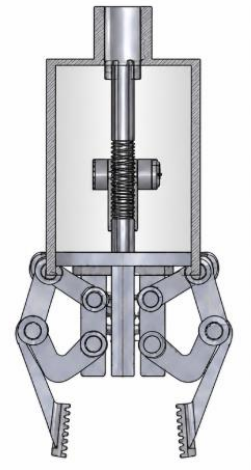
- mechanical complexity
- ease of assembly
- adaptability to different diameters and size
- time available for design and production.

Option 1

Demonstrates the inspiration of the piston-to-umbrella mechanism to grasp the component. This design mimics the mechanism in the figure below.

In this specific mechanism, a single slider moves an "endless screw" connected to the fingers. The fingers then move due to a system of mobile and fixed pins, allowing them to expand and contract similar to an umbrella.

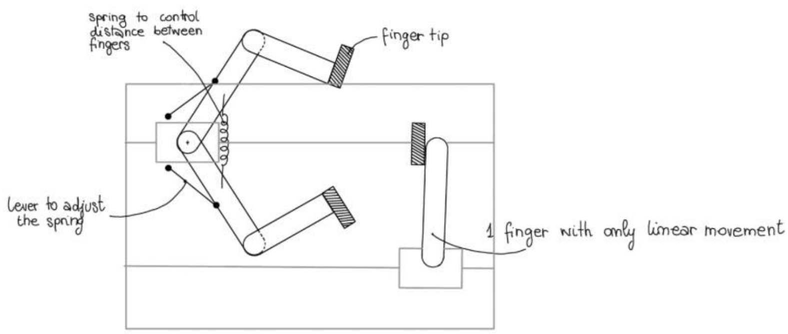




This is a viable option; however, when it comes to manufacturing and packaging, we have to consider the number of individual components involved and its difficulty in assembly and transporting. Also given the time constraints and the desire for a simple design for a simple task, we decided to simplify the design.

Option 2

The one finger gripper will be attached to the one-end motor component. While one of the other two fingers will be attached to the other end of the motor component. To adapt to the potential changing of the diameter of the component, we decided that we could adjust the spring stiffness prior to movement, and the adjustment would set the position.



As shown in the figure, one finger performs a purely linear movement, while the other two are connected via a spring mechanism. In this way, the distance between the fingers is controlled by a lever and a spring system. The idea was to adjust the spring stiffness prior to the movement to set the initial position, allowing the gripper to adapt to different component diameters.

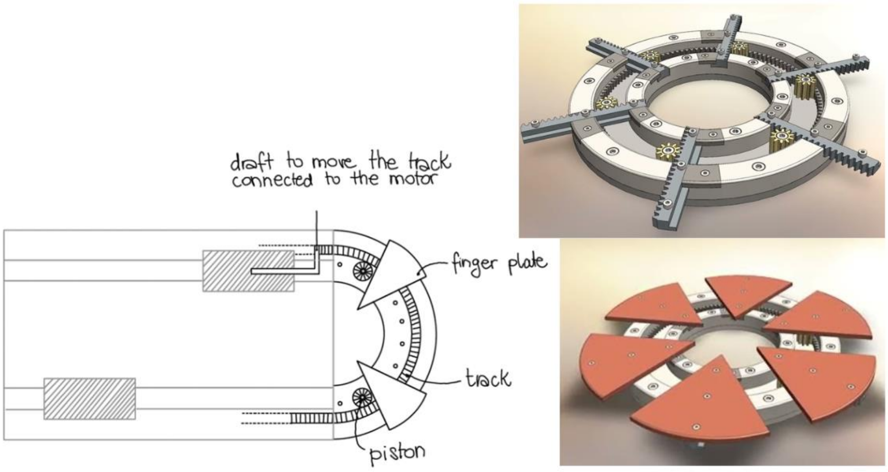
This is still appear complex, as we would have to consider the type of spring material and stiffness required to hold the fingers' position in place while lifting the load. Furthermore, it is necessary to select the correct type of stiffness; if the spring is too lenient, the component will slip; if it is too tight, it will damage the contact surface.

The benefits of this design are its low packaging and assembly costs. However, finding a spring with the exact stiffness required to balance both the grasping force and the weight of the load adds unnecessary complexity to the design phase.

Option 3

The third design option explores a more sophisticated approach inspired by an **"iris mechanism"**, as illustrated in the following image. In this concept, two of the fingers are mounted on a plate that moves along a circular track.

This circular movement is driven by a piston and a gear system (the "draft"), allowing the gripper to regulate its opening based on the component's diameter. Additionally, the finger plates can be manually positioned to set the initial distance.
Similar to the previous option, one slider controls a finger with pure linear motion, while the other slider is connected to the adaptor that moves the circular track.



While this mechanism offers a very high level of adaptability and a compact design, it presents significant challenges that led us to reconsider its execution.

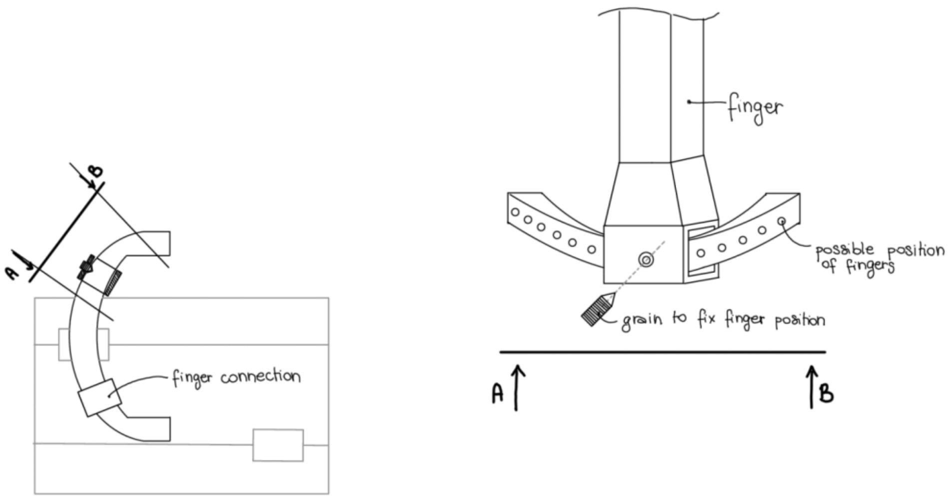
The main difficulty lies in the mechanical precision required for the assembly; designing the correct gears to achieve smooth meshing and the CAD model would be a timely task, and we would not be able to finish within the specified deadline. Another difficulty would be the maintenance and lubrication of the gears during its life cycle.

Option 4

After evaluating the previous iterations, we prioritized a solution that balances manufacturing simplicity with high operational reliability. We maintained the **modular slider architecture**: one slider controls a single finger, while the other moves a specialized support for the remaining two fingers.

The innovation involves using a "half-moon" guide to adjust the two-finger setup, as shown in the figure. This curved track allows the fingers to be positioned manually along a circular path before starting the operations. Once the optimal position is reached, the system securely locks the fingers in place using a set screw (grain) with pre-aligned positions to avoid any collision with the motor mount's flanges, ensuring more precise alignment and maintaining a permanent position. It also provides more compressive force to ensure stability when in contact with the component.

The **"half-moon" guide** solves the potential issues related to asymmetrical design. This implementation facilitates easier switching between the two fingers, enabling adjustments to the gripper for maintenance and component requirements. Furthermore, the guide features a series of small pre-drilled holes. These holes act as mechanical references, allowing the set screw to seat properly and ensuring that the finger position remains perfectly fixed even under the stress of lifting the load. This manual but precise adjustment makes the gripper highly adaptable and ready for different component variations without the need for complex internal mechanisms. It is also easy to assemble, package and carry out maintenance.



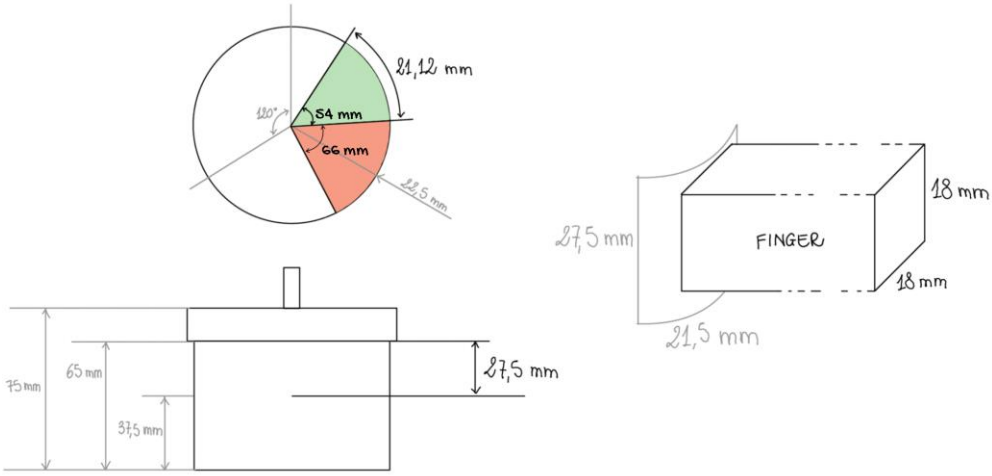
FINGER TIP

To define the correct finger dimensions, we analyzed the target component's geometry under the following assumptions:

- The three flanges are equidistant; there is an angular distance of 120° between them. For our calculation, we considered the largest diameter of **45 mm**, which results in a circumference of approximately **141.37 mm**.
- To ensure a safe operation and avoid any symmetry issues, we adopted a conservative approach by assuming the flanges occupy 55% of the total space. This means each "ungraspable" area covers an angle of approximately **66°**. Consequently, the available space for each finger is limited to an arc sector of **54°**. Using the arc formula *2*pi*r*(theta / 360)*, with a radius *r* of **22.5 mm**, we identified a maximum contact arc of **21.12 mm**.

Regarding the height of the fingertip, we considered the main body of the component (excluding the screw). By taking a reference point at **37.5 mm**, we calculated a safe vertical contact area of

27.5 mm². Based on these geometric constraints, we defined a standard **finger tip contact area** that fits within an **18 mm²** to maintain a safety margin from the flange edges and the ground.



Since the goal of our gripper is to be as adaptive as possible, we decided to provide the operator with a set of interchangeable fingers. This allows the gripper to handle different components by simply changing the finger size during the initial manual setup.

We established three categories based on a proportional linear reduction of **11.11%**, which corresponds to the step between one size to another with the respect of target diameters:

- **Size L:** Designed for the 45 mm diameter, using the **18 mm** square contact area as a baseline.
- **Size M:** Designed for the 40 mm diameter, derived by applying the 11.11% reduction, resulting in a contact area of approximately **16 mm**.
- **Size S:** Designed for the 35 mm diameter, further reduced to approximately **14.2 mm** to ensure a perfect fit for even smaller components or tighter spaces between flanges.

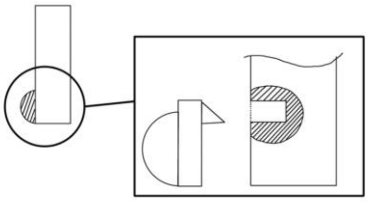
This modular approach ensures that the fingertip always maintains an optimal contact-to-surface ratio, preventing the gripper from being too bulky when working on smaller pieces.

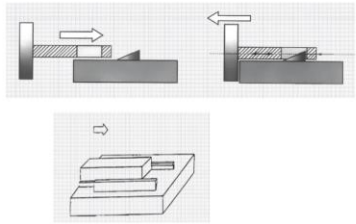
To prevent surface wear and tear and to improve the reliability of the grasp, we decided to incorporate a changeable **rubber pad** on each fingertip.

The attachment system is designed as a **sliding snap-fit** mechanism. The finger features a dedicated integrated guide, while the rubber pad is mounted onto a mobile plastic or metallic carrier designed to slide into this guide.

This configuration ensures that the rubber pad remains in the correct position throughout the trajectory. This prevents any accidental detachment.

Furthermore, this modular approach allows the operator to replace the rubber pads when they become worn out, without the need to disassemble the entire finger structure. By using this soft contact point, we also fulfill the requirement of protecting the painted surfaces and metallic surface of the component, as the rubber acts as a buffer that absorbs vibrations and prevents the metal fingers from chipping or scratching the component.





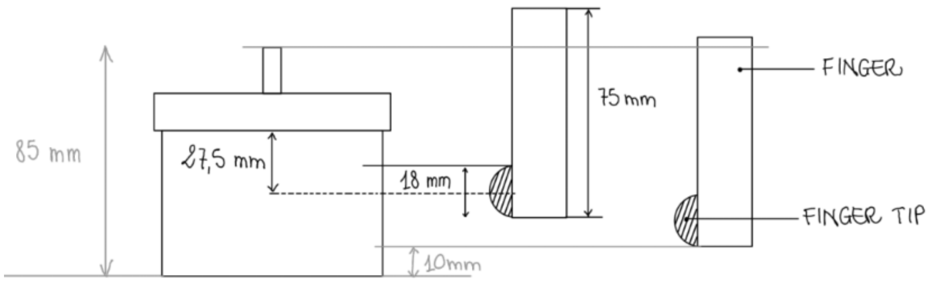
FINGER BODY

After defining the area for the fingertips, we established the necessary length for the finger body. The gripper must be able to reach the designated pick-up zone even on the highest version of the component.

To calculate the minimum required length, we considered the distance from the top of the component (**85 mm**) to the central pick-up point (**18.75 mm**), resulting in a minimum length of **66.25 mm**. However, to ensure maximum stability, we decided to set the finger length at **75 mm**.

This extended length offers two main advantages:

- **Lower Center of Gravity:** It allows the gripper to reach further down the cylindrical body, providing a more stable grasp and better balance during movement.
- **Safety Clearance:** It ensures that the main plate of the gripper and the Jodell motor remain at a safe distance from the top threaded screw, preventing any accidental contact or interference during the approach phase.



HALF-MOON CONNECTOR

Designing this component considers the constraints of the Jodell motor. We assume the center of the half-moon is aligned with one of the motors and with a diameter related to the largest diameter of the component (+/- 5mm for safety). This led to a radius of **27.5 mm.**

From this point, the ***next options*** *consider the follow changes:*

**Adaptability is now considered as the possibility for the gripper to adapt to the component. Therefore, the grippers setup is affected when in contact with the component.*

**For these options, one needs to consider the new assumptions regarding the geometry of the component. Which is as follows:*

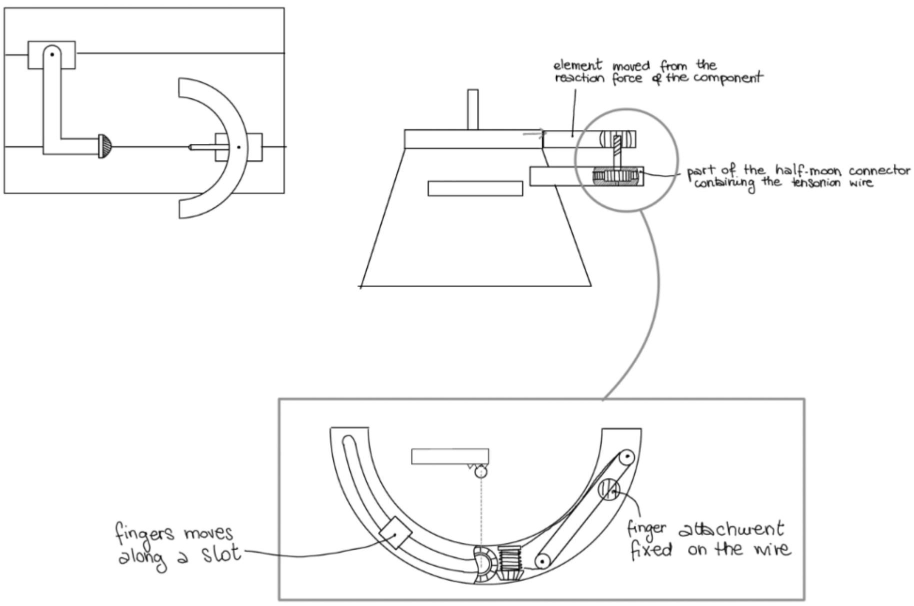
- *The distances for the allowable areas to touch are at a constant distance from the top of the component.*
- *The largest diameter of the component determines the size of our mechanism.*

Option 5

This option follows the previous one. To remove the manual setup and increase the compliance of the adaptability, we decided to add a new element to the half-moon component that ‘activates’ the movement of the two fingers. This element permits the fingers to adapt to the diameter of the component through a slider that starts from the outermost diameter and 'closes' around the component.

When this element touches the component’s surface, the reaction force exerted by the component causes the slider to move, translating linear motion to radial motion through an endless screw and proper gear assembly. The gear moves the tension wire, which allows the finger to slide along the half-moon slot. To allow this motion, a pulley system is integrated within the finger. For the pulley system to operate appropriately, a spring (which acts as a counterbalance) is attached to the other end of the pulley and to the end of the half-moon.

In this option the half-moon does not require the holes for the set screw anymore, but it needs a slot where the finger can slide along its length.



This idea automates the setup, as the movement is along a set trajectory. The mechanism allows the gripper to grab the component regardless of the diameter size.

However, this design requires careful consideration because the choice of materials and the setup of tension cables, springs, and pulley systems must effectively maintain the stresses applied to the finger. Additionally, the design adds to the complexity of assembly, maintenance and packaging, as it requires a setup of the tension cable, spring and pulley system. If implemented incorrectly, it would risk damaging the component.

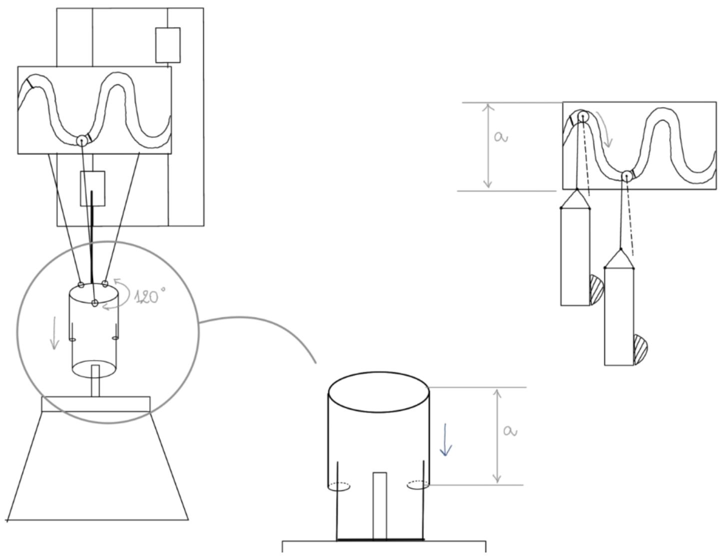
Option 6

We decided to have a synchronous movement of the three fingers that follow the trajectory of the half-moon. This approach allows us to design the mechanism without being limited by the diameter and height of the component itself, as it depends only on the angles.

The main part of this mechanism is the slotted guide, which is connected to the sliders of the Jodell motor. When the mechanism approaches the component, a section of it touches the top. Upon contact with the piece, a rigid bar descends along the guide, allowing the 'descent' of the fingers through the movement of the pin (connected to the rigid bar) within a slot. (As demonstrated in the figure).

The part of the mechanism that comes in contact with the component considers the maximum diameter of the upper screw of the target so as not to damage it.

The disadvantage of this mechanism concerns the slot itself; the geometry of the slot is indeed its main designing factor. The difficulty lies in the calculation of the exact geometry of the sinusoidal loop to ensure a smooth movement of the gripper.



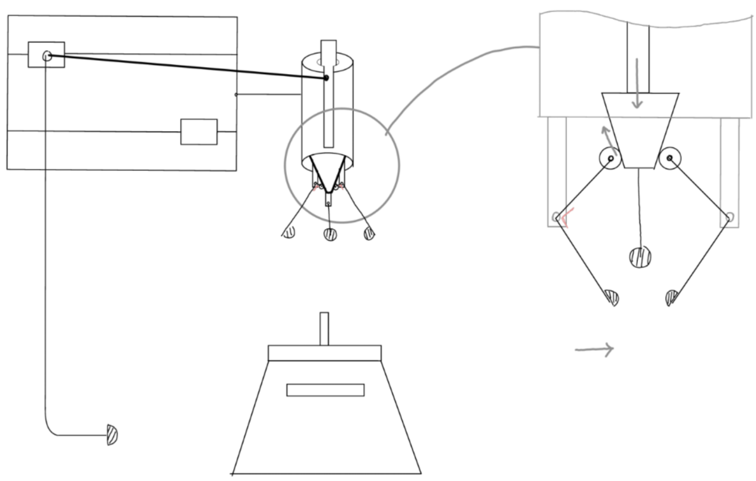
Option 7

Even in this case, the mechanism introduces a synchronous motion of the finger, considering only the 'touchable' areas of the component and thus allowing a more uniform adaptability without the need for manual setup.

The fingers ‘close’ on the demand of the linear motion of the conical element. The fingers roll across the conical element while pivoting about the rigid connection point.

The linear motion of the cone is activated by a rigid bar connected to the motor slider, which is also connected to a new bar that touches the component in the safe area. This permits the 'closure' of the finger with respect to the diameter size.

The additional element that has the first contact with the component is nothing more than a activation element; the same mechanism would work even by connecting the piston design directly to the motor slider.

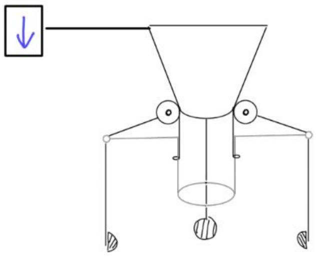


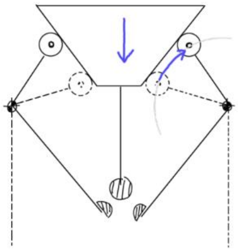
Option 8

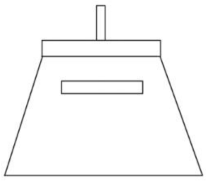
We considered optimising option 7 by employing the same approach as option 6.

The bottom cylinder, besides being the initial point of contact with the cylinder, functions as an activation for the pivoting of fingers. The fingers will rotate, similarly to option 7, about the conical component, which is firmly attached to the upper cylinder.

Like mechanism 6, the cylinders will slide over each other, but here, their movement will directly cause the fingers to wrap around the component. For mechanism 7, we used the conical component that facilitates finger mobility.







In this way, the adaptability is achieved by a simpler design.

Projects overview

Following is a summary outline of our proposals:

Opt.	Complexity	Reliability	Adaptability	Setup	Note
1	High	Medium	Low	No	Too many component to assembly
2	Medium	Low	Medium	Manual	Difficult balance between stiffness and the loads
3	High	High	High	Manual	Complex CAD and gear meshing
4	Low	High	High	Manual	Best balance of simplicity and modularity. However, it requires a long initial manual setup
5	High	Medium	High	Automatic	Design too complex for the presence of conical gears ,tension wires and spring
6	Medium	Medium	High	Automatic	Difficult to compute the exact geometry for the slot
7	Low	High	High	Automatic	Best balance of simplicity and automatic adaptability
8	Low	High	High	Automatic	Optimization of 7 option

Final design

After comparing the different options, the final design was developed by combining the most effective aspects of the previous solutions. The objective was to keep the adaptability of the automatic concepts while reducing the number of complex components and simplifying assembly. For this reason, the final solution is based on a three-finger architecture, with one finger moving linearly and two fingers guided through rigid bars and slots.

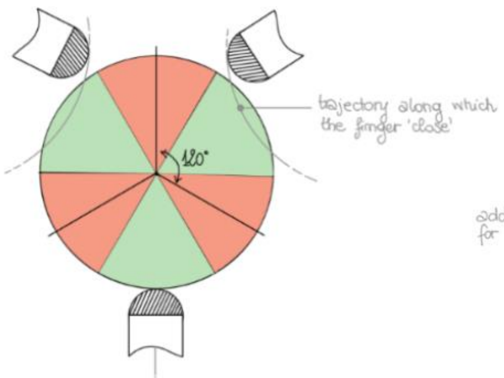
The scope of our gripper is to have a functional but simple design to allow an easy assembly without sacrificing adaptability. This design allows for the retrieval of components with varying diameters.

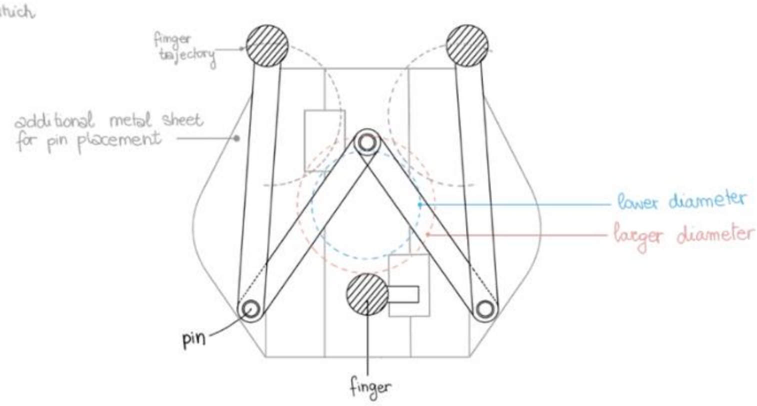
We opted for a gripper based on the contact angles dictated by the component. Since the petals/flanges are equidistant, a secure contact zone is available every 120°.

The largest top-mounted motor was utilised for the size of the fingers. The contact angle of the gripper's fingers is within the component's safe contact zone. Further considerations have been made to account for an offset of angled contact.

For this design, one finger is rigidly connected to one of the motors. The other two fingers are connected to the second motor through additional attachments (shown in the figures below). The other two fingers follow a curved trajectory. This is achieved via a pin connection on the metal sheets and a pin and slot connection on the second motor. We utilised a pin screw to attach the fingers to the slider and the sheet metal. The edges of the bent sheet metal were rounded, reducing the overall unused material. Moreover, to avoid any tilting and guarantee alignment of the sheet metal with the Jodell edges, the full length of the motor and both screws and set holes were used to secure the sheet metal to the motor.

The fingers were placed at an offset to be in the centre of the motor. This is to ensure the gripper can adjust to different sized diameters. This led to an additional metal-guided railing being added to the motor to assist in the stability of the fingers. Considering the reduced dimensions of the component and the small additional radius, the moment caused by hanging mass off the motor can be neglected.



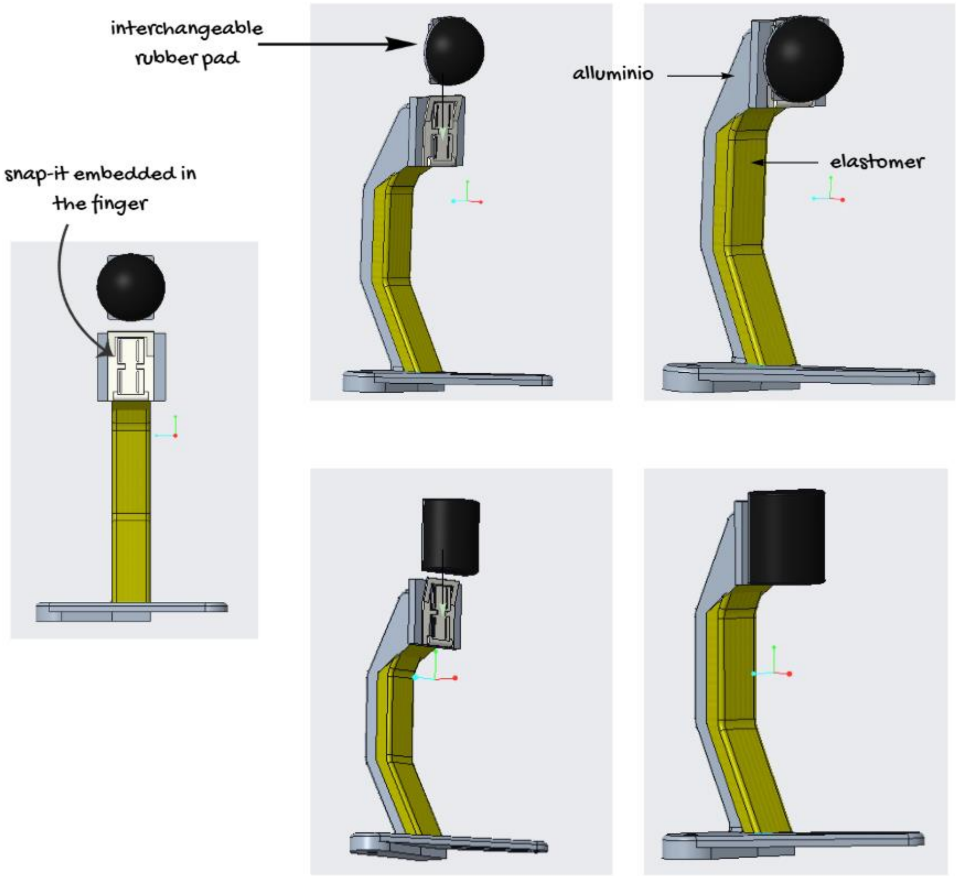


This design should guarantee a compromise between adaptability, development difficulty and functionality.

Fingers

To ensure stable pickup, the fingers include an external metal component that supports the weight of the object and an interior silicone/elastomer element that enhances stability.

For the height of the fingers, we considered the largest diameter to define the minimum possible height and avoid contact with the upper surface and the screw. Considering additional millimetres, we define the height of the fingers as **72.82 mm** (considering the additional rubber attachment). We added washer between the sliding plates holding the two fingers to reduce wear and tear. To avoid unused material, the dimensions of where the single finger is attached were reduced. To counteract the bending moment felt when force is applied to the singular finger when in contact with the component, a second screw behind the finger was added.



2-Fingers Dynamics

Starting from the diameter of the component, we considered both the smallest and largest configurations. For the largest diameter, we added 3 mm as a safety margin. These diameters were used to define the safe areas that the fingers have to reach during the closing movement.

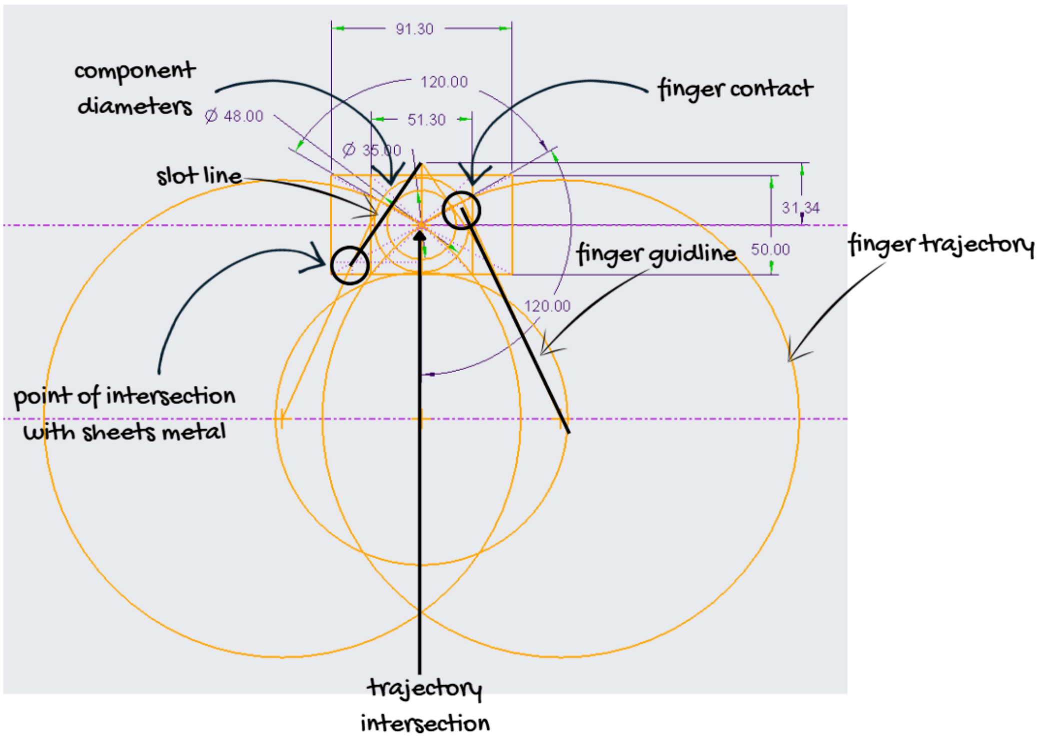
Since the component has three petals positioned at about 120°, the fingers also had to keep a similar angular trajectory. In this way, the gripper can avoid the interference with the flanges and touch only the safe areas of the component.

To define the synchronous radial movement of the two lateral fingers, we traced the possible trajectories that the fingertips had to follow. The intersection between these trajectories identifies the area in which the fingers can move while remaining inside the safe contact region. For this reason, the two large circles represent the theoretical trajectory followed by the fingertips during the linear motion of the sliders.

To define the geometry of the slot, we considered the point where the pin screw will be attached to the slider and the intersection of the trajectory line from the centre of the larger trajectory circles. This reference ensures that the two fingers have the ability to cover all the possible dimensions of the component.

The length of where the point of contact of the finger should be is determined by the intersection of the trajectory line from the centre of the larger trajectory circles and the larger component diameter.

Due to the arc-finger design, the contact point and the base of the finger were not aligned, which is why the finger levers are larger than expected. The fingers were also angled to be parallel to the line of the slot to ensure, as much as possible, a contact point tangent to the rubber attachment.

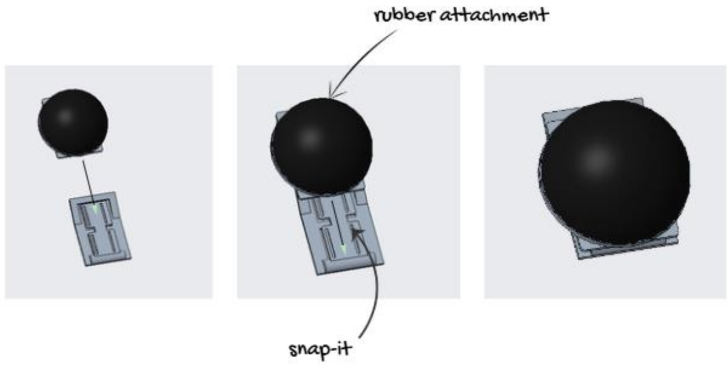


Fingers tip

At the fingertips, a snap-fit mechanism is used to secure a rubber attachment. The rubber contact with the component ensures a firm connection. We opted for a snap-fit mechanism to handle wear and tear issues. Upon continuous use of the rubber, the user can effortlessly substitute that component without the need to disassemble other elements of the gripper.

We designed two types of rubber attachments: two **spherical** ones and one **cylindrical** one. The spherical attachments were attached to the two fingers and acted as contact points. However, since the component is small, we decided to also include a cylindrical attachment that is attached to a single finger. The design of the cylinder reduces the risk of tilting or bending movements. It also assists in stability because the full surface area of the two-finger rubber attachment will not always be in contact with the component, so having a larger surface area that is always in contact provides better stability.

The snap-fit mechanism is larger than the final contact area of the finger, with the rubber attachment covers an area of **18 mm^2**, following the same reasoning described in **Option 4** in the **Finger Tip** paragraph. It is secured to the part with 4 countersunk screws, to ensure ease of assembly and stability.



Requirement verification

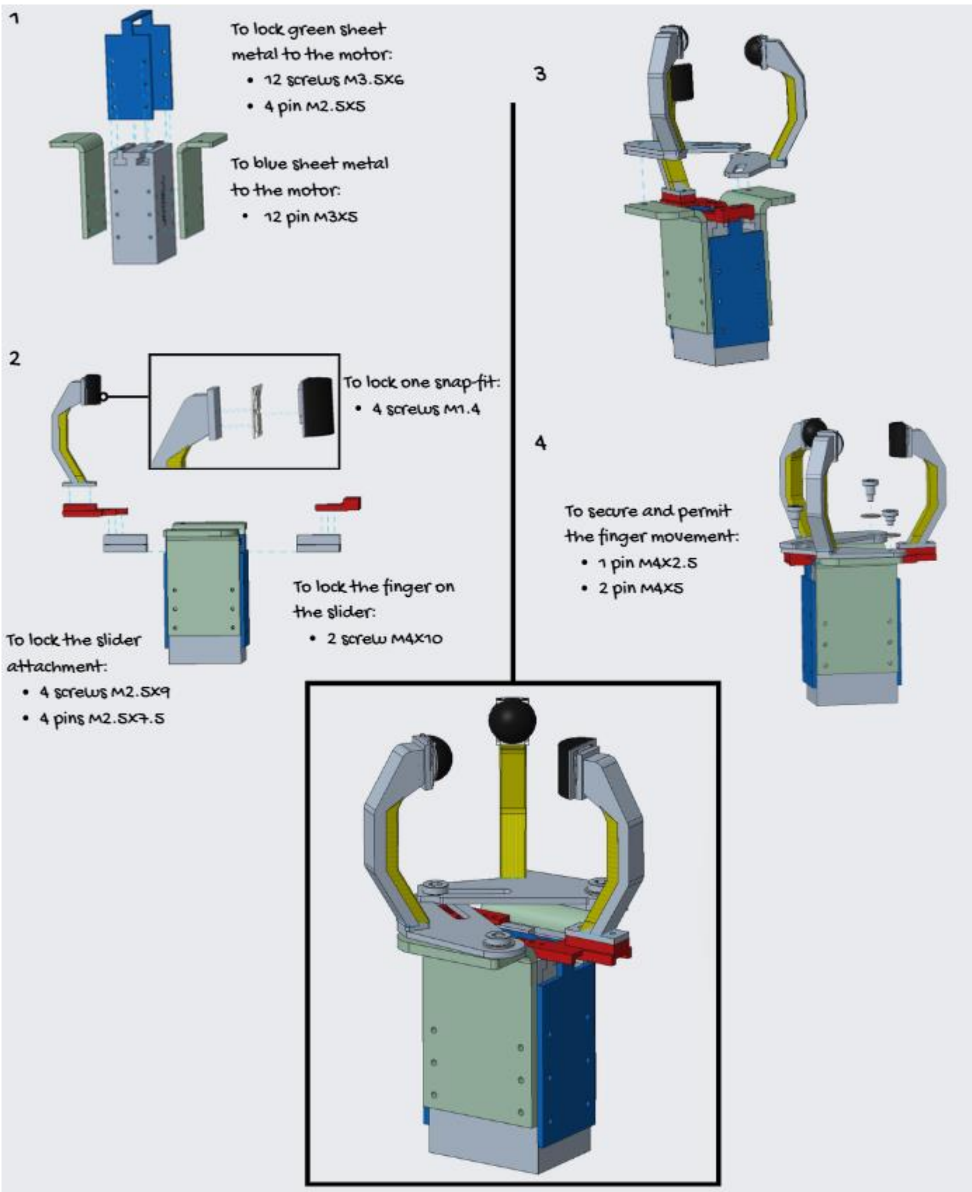
In this section we will study how the solution presented satisfies the given requirements:

Requirement	Description	How it is satisfied
Safe grasping	The gripper must avoid the threaded screw, bolt sections and interference with the flanges	The height of the fingers allows the gripper to reach the lower cylindrical surface, while the 120° movement of the fingers allows them to avoid the flanges
Adaptability	The gripper must be able to grasp components with different diameters	The finger movement is based on the angular position of the safe areas. Since the fingers keep an angle of about 120°, the gripper can adapt to different diameters
Stability	The component must remain stable during the pick-up operation	The use of two spherical contacts and one cylindrical contact improves stability and reduces the risk of tilting
Compliant grasping	The gripper should provide a softer and more adaptable contact with the component surface	The elastomer part on the fingers improves compliance along the component surface, while the rubber attachments help the fingers adapt to the local geometry during the pick-up
Easy assembly	The design should be simple and feasible to assemble	The reduced number of components and the modular finger structure simplify the assembly process

Maintenance	Wearable parts should be easy to replace	The rubber attachments are replaceable and can be changed without disassembling the whole gripper
Compatibility	The mechanism should be compatible with the Jodell motor and its linear sliders.	One finger is connected to one slider and performs a linear movement, while the other two fingers are moved through rigid bars and slots connected to the second slider of the motor

Assembly

The assembly begins with the Jodell motor's body, incorporating its two main support metal sheets. The singular finger is mounted to one slider, whereas the two lateral fingers are linked to the second slider via rigid bars, pins, and slots. The rubber attachments are fitted into the snap-fit guides located at the ends of each finger. This procedure facilitates modular assembly of the gripper and ensures efficient maintenance.

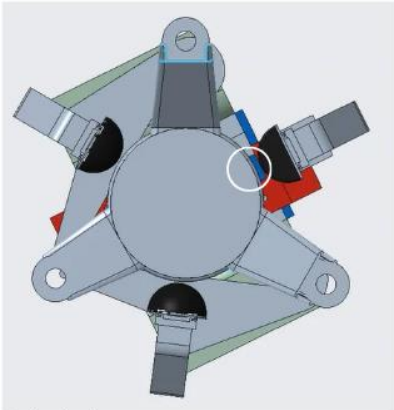


Conclusion

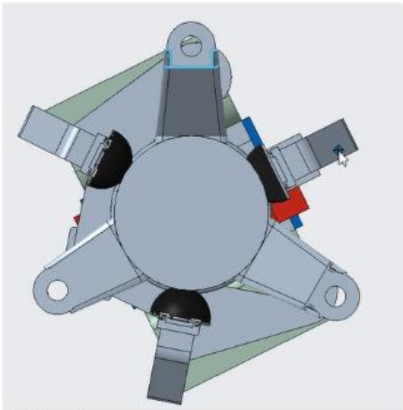
To conclude, the proposed gripper represents a good compromise between functionality, adaptability and manufacturing simplicity. The final solution satisfies the main requirements defined at the beginning of the project, allowing the component to be grasped from the safe lower cylindrical surface while avoiding the restricted areas.

The three-finger architecture, together with the replaceable rubber attachments and the simplified mechanical structure, supports a stable pick-up, easy maintenance and a feasible assembly process. For these reasons, the design can be considered suitable for the required task.

Finally, the final CAD test confirms that the gripper can close correctly, as seen in the test below, conducted with the component with the largest diameter. The fingers remain aligned with the safe contact areas and avoid interference with the flanges, validating the design approach used.



Test start:
the gripper has enough tolerance prior to motion



Test end:
the rubber ensure enough pressure on the component to guarantee better pick up

Possible future adjustments for Manufacturing

The following are possible future design improvements to reduce costs and increase ease of manufacturing:

- Modify and make the fingers in sheet metal rather than steel as in the current version.
- Perform cuts to further reduce material at various points of the design, such as in the lateral supports.
- Add laser cuts also in the bases of the two rotating fingers to facilitate the assembly of the part.
- Add a metal notch inside the rubber of the fingertips to improve the stability and firmness of the grip.